Area 3: Mechanical Attachments

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

Contents

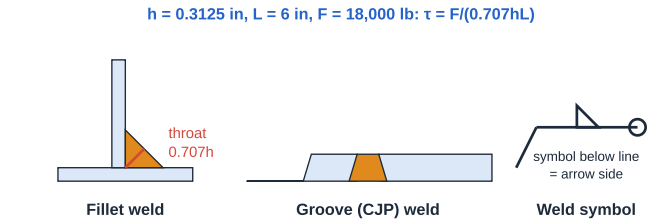
3.A Bonds (Welding, Brazing, Adhesives)
3.B Non-threaded Fasteners
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3.A Bonds (Welding, Brazing, Adhesives)

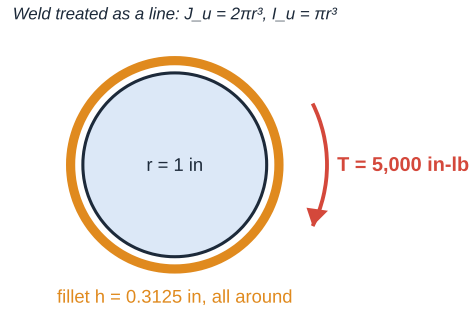
1. Key concepts

- Fillet weld strength is set by the throat: t = 0.707h for an equal-leg fillet. Shear stress τ = F/(0.707hL).
- Weld groups under torsion or bending: treat the weld as a line. Find the unit properties J_u and I_u, then J = 0.707hJ_u. Add the primary (direct) and secondary (moment) shear as vectors.
- Weld symbols (AWS A2.4): a symbol below the reference line means arrow side; above means other side. A flag means field weld, a circle means all around, and the tail carries the process or spec.
- Adhesives are strong in shear and weak in peel. Lap joint: τ_avg = F/(bL).

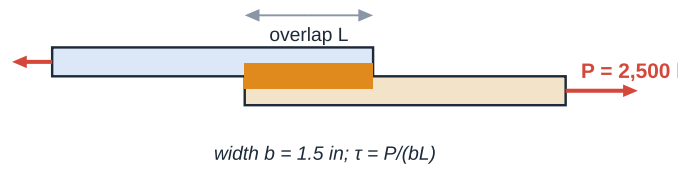
Diagrams



Fillet vs groove welds; throat = 0.707h.



Round bar welded all around: treat the weld as a line.



Adhesive single-lap joint.

2. Essential formulas

Name	Formula	Variables / units
Fillet throat stress	$\tau = F/(0.707hL)$	
Weld torsion	$\tau'' = Tr/J\ ;\ J = 0.707h \cdot J_u$	circle $J_u = 2\pi r^3$
Weld bending	$\tau = Mc/I\ ;\ I = 0.707h \cdot I_u$	circle $I_u = \pi r^3$
Adhesive lap	$\tau = F/(bL)$	

3. Handbook search keywords

fillet weld throat · weld group polar moment · welding symbols · adhesive lap joint

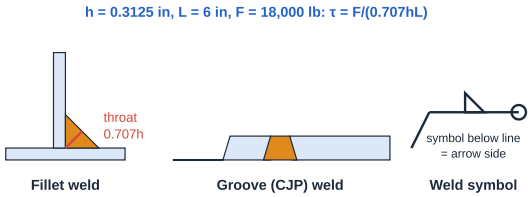
4. Worked examples

Example 1: Parallel fillet welds
Two 0.3125 in fillet welds, each 6 in long, carry 18,000 lb in shear. Find τ.
Handbook search: **fillet weld throat**

1. Throat area = 2(0.707)(0.3125)(6) = 2.651 in².
2. τ = 18,000/2.651 = 6,789 psi.

Answer: 6,789 psi

Check your answer: Throat area = 0.707hL per weld; with two welds, double it.

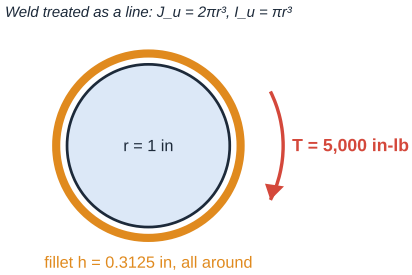


Try it yourself: Weld in torsion
A Ø2 in bar (r = 1 in) is welded all around with a 0.3125 in fillet and carries 5,000 in-lb. Find the throat shear.
Handbook search: **weld as a line**

1. Unit polar moment J_u = 2πr³ = 6.283 in³.
2. τ = Tr/(0.707h·J_u) = 3,602 psi.

Answer: 3,602 psi

Check your answer: Same answer from τ = T/(0.707h·2πr²) = 3,602 psi. ✔

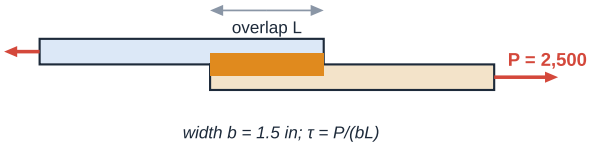


Example 3: Adhesive overlap length
Strips 1.5 in wide must carry 2,500 lb; the adhesive allows 1,500 psi. Find the minimum overlap.
Handbook search: **adhesive lap joint**

1. Area needed = 2,500/1,500 = 1.667 in².
2. Overlap = area/width = 1.111 in.

Answer: 1.11 in

Check your answer: Real lap joints carry peak stress at the ends. Treat the average-stress answer as a minimum.



5. Common traps

- Using the leg h instead of the throat 0.707h.
- Adding the primary and secondary shear as scalars instead of vectors.
- Reversing arrow side and other side on the symbol.

3.A Bonds (Welding, Brazing, Adhesives): one-page cheat summary

- ► $t = 0.707h$
- ► $\tau = F/(0.707h\Sigma L)$
- ► $J = 0.707h \cdot J_u$
- ► Below line = arrow side

Formulas

Fillet throat stress	$\tau = F/(0.707hL)$
Weld torsion	$\tau'' = Tr/J$; $J = 0.707h \cdot J_u$
Weld bending	$\tau = Mc/I$; $I = 0.707h \cdot I_u$
Adhesive lap	$\tau = F/(bL)$

Search keywords

fillet weld throat, weld group polar moment, welding symbols, adhesive lap joint

Top traps

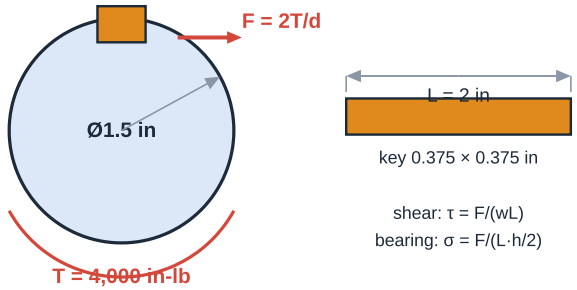
- Using the leg h instead of the throat 0.707h.
- Adding the primary and secondary shear as scalars instead of vectors.
- Reversing arrow side and other side on the symbol.

3.B Non-threaded Fasteners

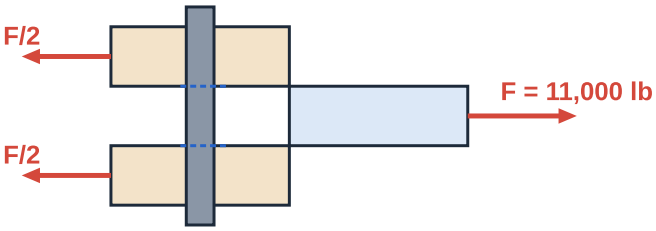
1. Key concepts

- Pins, rivets and keys: check shear (single or double), bearing (crushing, σ_b = F/(dt)), and net-section tension of the plate, σ = F/((w − d)t).
- Keys: the force at the shaft surface is F = 2T/d. Shear τ = F/(wL); bearing on half the height σ = F/(L·h/2).
- Retaining rings, roll pins and shoulders locate parts axially.

Diagrams

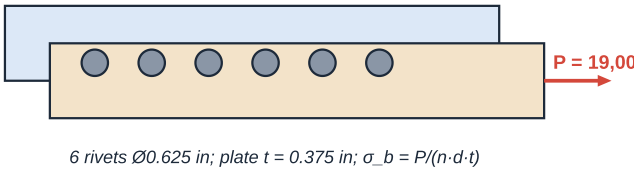


Square key: shear on w·L, bearing on (h/2)·L.



Ø0.75 in pin, double shear: τ = F/(2A)

Clevis pin in double shear.



6 rivets Ø0.625 in; plate t = 0.375 in; σ_b = P/(n·d·t)

Riveted lap joint: shear, bearing, net tension.

2. Essential formulas

Name	Formula	Variables / units
Key force	$F = 2T/d$	
Key shear	$\tau = F/(wL)$	
Key bearing	$\sigma = 2F/(hL)$	square key h = w
Bearing on plate	$\sigma_b = F/(n \cdot d \cdot t)$	
Net section	$\sigma = F/((w - \Sigma d)t)$	

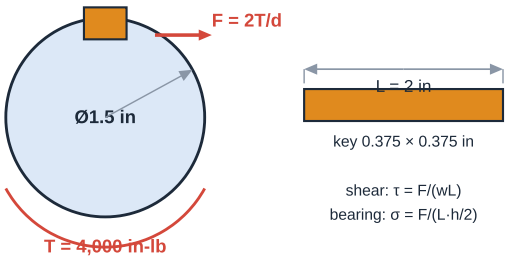
3. Handbook search keywords

key shear · rivet bearing stress · pin double shear · net section

4. Worked examples

Example 1: Square key length
A 0.375 in square key on a Ø1.5 in shaft carries 4,000 in-lb. Key Sy = 60 ksi, n = 2.5, Ssy = 0.577Sy. Find the minimum L.

Handbook search: **key design**



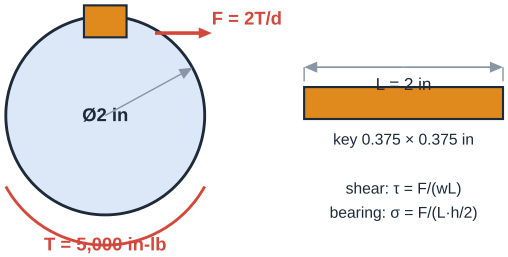
1. F = 2(4,000)/1.5 = 5,333 lb.
2. Shear: L = Fn/(0.577·Sy·w) = 1.027 in.
3. Bearing: L = Fn/(Sy·w/2) = 1.185 in. Bearing governs.

Answer: 1.19 in

Check your answer: Check both shear and crushing. Here crushing gives the same length because the key is square and Ssy ≈ Sy/2.

Try it yourself: Key shear stress
A Ø2.0 in shaft transmits 5,000 in-lb through a 0.375 in square key, 2 in long. Find τ.

Handbook search: **key shear**



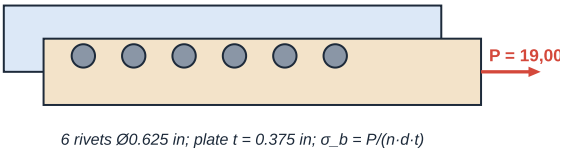
1. F = 2T/d = 5000 lb.
2. τ = F/(wL) = 5,000/(0.375 × 2) = 6,667 psi.

Answer: 6,667 psi

Check your answer: Bearing stress on half the height is 2× this (13,333 psi) for a square key.

Example 3: Rivet bearing stress
6 rivets, Ø0.625 in, through 0.375 in plates carry 19,000 lb. Find the plate bearing stress.

Handbook search: **bearing stress**



6 rivets Ø0.625 in; plate t = 0.375 in; σ_b = P/(n·d·t)

1. Bearing area = n·d·t = 6 × 0.625 × 0.375 = 1.406 in².
2. σ_b = 19,000/1.406 = 13,511 psi.

Answer: 13,511 psi

Check your answer: Bearing uses the projected area d·t, not πd²/4.

5. Common traps

- Using the diameter instead of the radius in F = T/r.
- Forgetting the bearing check on half the key height.

3.B Non-threaded Fasteners: one-page cheat summary

- ► $F = 2T/d$
- ► $\tau = F/wL$
- ► $\sigma_b = 2F/hL$
- ► $net = F/((w - d)t)$

Formulas

Key force	$F = 2T/d$
Key shear	$\tau = F/(wL)$
Key bearing	$\sigma = 2F/(hL)$
Bearing on plate	$\sigma_b = F/(n \cdot d \cdot t)$
Net section	$\sigma = F/((w - \Sigma d)t)$

Search keywords

key shear, rivet bearing stress, pin double shear, net section

Top traps

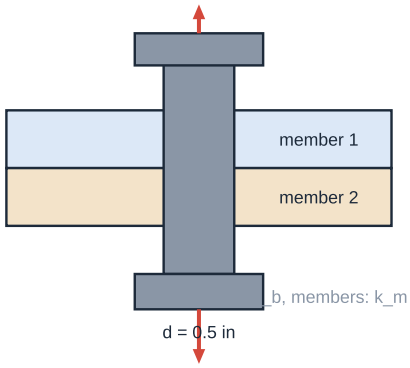
- Using the diameter instead of the radius in F = T/r.
- Forgetting the bearing check on half the key height.

3.C Threaded Fasteners

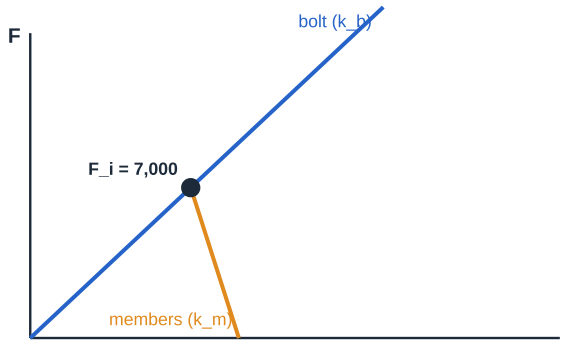
1. Key concepts

- Preload: $F_i = 0.75 A_t S_p$ for a reusable joint, $0.90 A_t S_p$ for a permanent one. Torque $T = K F_i d$ with $K \approx 0.2$ dry or 0.15 lubricated.
- Joint constant $C = k_b/(k_b + k_m)$. The bolt picks up $C \cdot P$ of the external load and the members lose $(1 - C)P$. The joint separates at $P_0 = F_i/(1 - C)$.
- Bolt fatigue: $\sigma_a = CP/(2A_t)$ and $\sigma_m = \sigma_a + F_i/A_t$. Use Goodman with the preload line.
- Eccentric bolt groups: primary shear P/n plus secondary shear $F = Mr/\Sigma r^2$, added as vectors.

Diagrams



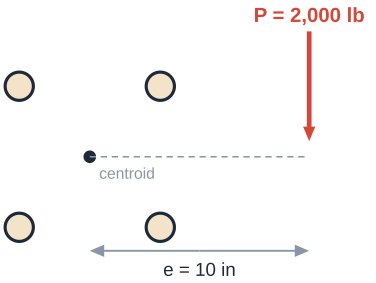
Bolted joint: bolt in tension, members in compression.



C = 0.231; separation at $P_0 = F_i/(1-C)$ = 9,103

bolt load at P=3,000: $F_i + CP$ = 7,693

Joint diagram: preload, load sharing and separation.



F = P/n (direct) + M·r/Σr² (moment), added as vectors

Eccentric bolt group: direct plus torsional shear.

2. Essential formulas

Name	Formula	Variables / units
Preload	$F_i = 0.75 A_t S_p$ (reusable)	
Torque	$T = K F_i d$	
Joint constant	$C = k_b/(k_b + k_m)$	
Bolt load	$F_b = F_i + CP$	
Separation	$P_0 = F_i/(1 - C)$	
Eccentric group	$F'' = Mr_i/\Sigma r^2$	

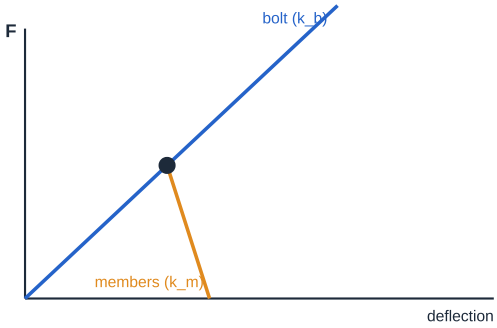
3. Handbook search keywords

bolt preload | torque coefficient | joint stiffness constant | bolted joint separation | eccentric bolt group

4. Worked examples

Example 1: Preload, torque and separation
A 1/2-13 UNC grade 5 bolt (A_t = 0.1419 in², S_p = 85 ksi), reusable, K = 0.2, k_b = 3 and k_m = 10 (×10⁶ lb/in). Find F_i, T, the bolt load at P = 2,000 lb, and the separation load.
Handbook search: **joint stiffness constant**

1. $F_i = 0.75(0.1419)(85{,}000) = 9{,}046$ lb.
2. $T = 0.2(9{,}046)(0.5) = 905$ in-lb.
3. $C = 3/13 = 0.231$; $F_b = 9{,}046 + 0.231(2{,}000) = 9{,}508$ lb.
4. $P_0 = 9{,}046/(1 - 0.231) = 11{,}760$ lb.

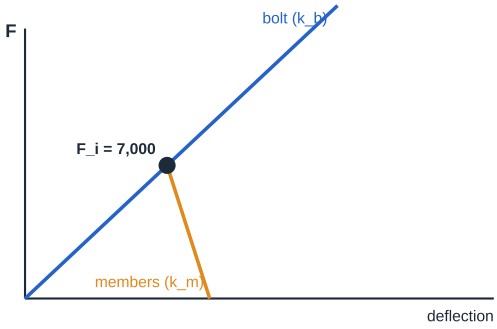


Answer: $F_i \approx 9{,}046$ lb; $T \approx 905$ in-lb; $P_0 \approx 11{,}760$ lb

Check your answer: Bolt load must rise only by C·P, and C is usually 0.2–0.3 for steel members.

Try it yourself: Bolt load with external load
F_i = 7,000 lb, k_b = 3, k_m = 10 (×10⁶ lb/in), external P = 3,000 lb. Find the bolt tension.
Handbook search: **bolted joint stiffness**

1. $C = k_b/(k_b + k_m) = 0.2308$.
2. $F_b = F_i + CP = 7{,}000 + 0.2308 \times 3{,}000 = 7{,}692$ lb.

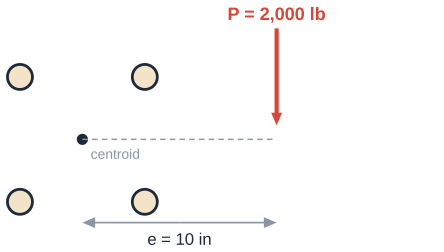


Answer: 7,692 lb

Check your answer: Members lose (1 − C)P = 2,308 lb of clamp. They have not separated because that is less than F_i.

Example 3: Eccentric bolt group
4 bolts on a 3 in square, 2,000 lb acting 10 in from the centroid. Find the worst bolt load.
Handbook search: **eccentric loading bolt group**

1. Direct: P/4 = 500 lb each.
2. $r = \sqrt{(1.5^2 + 1.5^2)} = 2.121$ in; torsional $F_t = Pe \cdot r/\Sigma r^2 = 2{,}357$ lb.
3. At the worst bolt the torsional force is at 45° and adds to the direct shear.
4. $R = \sqrt{(1{,}667^2 + 2{,}167^2)} = 2{,}734$ lb.



F = P/n (direct) + M·r/Σr² (moment), added as vectors

Answer: 2,734 lb

Check your answer: The worst bolt must carry more than both P/4 = 500 lb and F_t alone.

5. Common traps

- Giving the bolt the whole external load (it only takes C·P).
- Using the major-diameter area instead of A_t.
- Using 0.90 preload when the joint is reusable.

3.C Threaded Fasteners: one-page cheat summary

- ► $F_i = 0.75A_tS_p$
- ► $T = KF_id$
- ► $C = k_b/(k_b + k_m)$
- ► $F_b = F_i + CP$
- ► $P_0 = F_i/(1 - C)$

Formulas

Preload	$F_i = 0.75A_tS_p$ (reusable)
Torque	$T = K{\cdot}F_i{\cdot}d$
Joint constant	$C = k_b/(k_b + k_m)$
Bolt load	$F_b = F_i + CP$
Separation	$P_0 = F_i/(1 - C)$
Eccentric group	$F'' = Mr_i/\Sigma r^2$

Search keywords

bolt preload, torque coefficient, joint stiffness constant, bolted joint separation, eccentric bolt group

Top traps

- Giving the bolt the whole external load (it only takes C·P).
- Using the major-diameter area instead of A_t.
- Using 0.90 preload when the joint is reusable.